MSG No. 128.409 P_C4/mnc [Type IV, tetragonal]

Table 1: Wyckoff site: `4a`, site symmetry: `4'2'2`

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[1,6,7,8,18,19,20,21]`
2	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[2,3,4,5,17,22,23,24]`
3	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[9,14,15,16,26,27,28,29]`
4	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[10,11,12,13,25,30,31,32]`

Table 2: Wyckoff site: `4b`, site symmetry: `4'/m..`

No.	position	mapping
1	$\left[0, \frac{1}{2}, 0\right]$	`[1,6,9,14,18,19,26,27]`
2	$\left[\frac{1}{2}, 0, 0\right]$	`[2,3,10,11,17,22,25,30]`
3	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[4,5,12,13,23,24,31,32]`
4	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[7,8,15,16,20,21,28,29]`

Table 3: Wyckoff site: `4c`, site symmetry: `42'2'`

No.	position	mapping
1	$\left[0, 0, \frac{1}{4}\right]$	`[1,2,3,6,20,21,23,24]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[4,5,7,8,17,18,19,22]`
3	$\left[0, 0, \frac{3}{4}\right]$	`[9,10,11,14,28,29,31,32]`
4	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[12,13,15,16,25,26,27,30]`

Table 4: Wyckoff site: `4d`, site symmetry: `4/m..`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,3,6,9,10,11,14]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[4,5,7,8,12,13,15,16]`
3	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[17,18,19,22,25,26,27,30]`
4	$\left[0, 0, \frac{1}{2}\right]$	`[20,21,23,24,28,29,31,32]`

Table 5: Wyckoff site: `8e`, site symmetry: `2'/m..`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	`[1,14,22,25]`
2	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[2,11,19,26]`
3	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[3,10,18,27]`
4	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[4,13,21,28]`
5	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[5,12,20,29]`
6	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	`[6,9,17,30]`
7	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[7,16,24,31]`

8	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	[8,15,23,32]

Table 6: Wyckoff site: 8f, site symmetry: 2.22

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[1,8,22,23]
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[2,4,19,21]
3	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[3,5,18,20]
4	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[6,7,17,24]
5	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[9,16,30,31]
6	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[10,12,27,29]
7	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[11,13,26,28]
8	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[14,15,25,32]

Table 7: Wyckoff site: 8g, site symmetry: 4'..

No.	position	mapping
1	$\left[0, \frac{1}{2}, z\right]$	[1,6,18,19]
2	$\left[\frac{1}{2}, 0, z\right]$	[2,3,17,22]
3	$\left[\frac{1}{2}, 0, \frac{1}{2} - z\right]$	[4,5,23,24]
4	$\left[0, \frac{1}{2}, \frac{1}{2} - z\right]$	[7,8,20,21]
5	$\left[0, \frac{1}{2}, -z\right]$	[9,14,26,27]
6	$\left[\frac{1}{2}, 0, -z\right]$	[10,11,25,30]
7	$\left[\frac{1}{2}, 0, z + \frac{1}{2}\right]$	[12,13,31,32]
8	$\left[0, \frac{1}{2}, z + \frac{1}{2}\right]$	[15,16,28,29]

Table 8: Wyckoff site: 8h, site symmetry: 4..

No.	position	mapping
1	$[0, 0, z]$	[1,2,3,6]
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2} - z\right]$	[4,5,7,8]
3	$[0, 0, -z]$	[9,10,11,14]
4	$\left[\frac{1}{2}, \frac{1}{2}, z + \frac{1}{2}\right]$	[12,13,15,16]
5	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	[17,18,19,22]
6	$\left[0, 0, \frac{1}{2} - z\right]$	[20,21,23,24]
7	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	[25,26,27,30]
8	$\left[0, 0, z + \frac{1}{2}\right]$	[28,29,31,32]

Table 9: Wyckoff site: 16i, site symmetry: 2'..

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{3}{4}, z\right]$	[1,22]
2	$\left[\frac{1}{4}, \frac{3}{4}, z\right]$	[2,19]
3	$\left[\frac{3}{4}, \frac{1}{4}, z\right]$	[3,18]

continued ...

Table 9

No.	position	mapping
4	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{2}-z\right]$	[4,21]
5	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{2}-z\right]$	[5,20]
6	$\left[\frac{1}{4},\ \frac{1}{4},\ z\right]$	[6,17]
7	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{2}-z\right]$	[7,24]
8	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{2}-z\right]$	[8,23]
9	$\left[\frac{1}{4},\ \frac{1}{4},\ -z\right]$	[9,30]
10	$\left[\frac{3}{4},\ \frac{1}{4},\ -z\right]$	[10,27]
11	$\left[\frac{1}{4},\ \frac{3}{4},\ -z\right]$	[11,26]
12	$\left[\frac{3}{4},\ \frac{1}{4},\ z+\frac{1}{2}\right]$	[12,29]
13	$\left[\frac{1}{4},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	[13,28]
14	$\left[\frac{3}{4},\ \frac{3}{4},\ -z\right]$	[14,25]
15	$\left[\frac{3}{4},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	[15,32]
16	$\left[\frac{1}{4},\ \frac{1}{4},\ z+\frac{1}{2}\right]$	[16,31]

Table 10: Wyckoff site: 16j, site symmetry: .2'.

No.	position	mapping
1	$\left[0,\ y,\ \frac{1}{4}\right]$	[1,21]
2	$\left[-y,\ 0,\ \frac{1}{4}\right]$	[2,24]
3	$\left[y,\ 0,\ \frac{1}{4}\right]$	[3,23]
4	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{4}\right]$	[4,22]
5	$\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{4}\right]$	[5,17]
6	$\left[0,\ -y,\ \frac{1}{4}\right]$	[6,20]
7	$\left[y+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	[7,19]
8	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ \frac{1}{4}\right]$	[8,18]
9	$\left[0,\ -y,\ \frac{3}{4}\right]$	[9,29]
10	$\left[y,\ 0,\ \frac{3}{4}\right]$	[10,32]
11	$\left[-y,\ 0,\ \frac{3}{4}\right]$	[11,31]
12	$\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{3}{4}\right]$	[12,30]
13	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	[13,25]
14	$\left[0,\ y,\ \frac{3}{4}\right]$	[14,28]
15	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ \frac{3}{4}\right]$	[15,27]
16	$\left[y+\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	[16,26]

Table 11: Wyckoff site: 16k, site symmetry: ..2

No.	position	mapping
1	$\left[x,\ x+\frac{1}{2},\ \frac{1}{4}\right]$	[1,7]
2	$\left[\frac{1}{2}-x,\ x,\ \frac{1}{4}\right]$	[2,5]
3	$\left[x+\frac{1}{2},\ -x,\ \frac{1}{4}\right]$	[3,4]
4	$\left[-x,\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	[6,8]
5	$\left[-x,\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	[9,15]
6	$\left[x+\frac{1}{2},\ -x,\ \frac{3}{4}\right]$	[10,13]
7	$\left[\frac{1}{2}-x,\ x,\ \frac{3}{4}\right]$	[11,12]

continued ...

Table 11

No.	position	mapping
8	$\left[x,\, x+\frac{1}{2},\, \frac{3}{4}\right]$	[14,16]
9	$\left[x+\frac{1}{2},\, x,\, \frac{1}{4}\right]$	[17,23]
10	$\left[-x,\, x+\frac{1}{2},\, \frac{1}{4}\right]$	[18,21]
11	$\left[x,\, \frac{1}{2}-x,\, \frac{1}{4}\right]$	[19,20]
12	$\left[\frac{1}{2}-x,\, -x,\, \frac{1}{4}\right]$	[22,24]
13	$\left[\frac{1}{2}-x,\, -x,\, \frac{3}{4}\right]$	[25,31]
14	$\left[x,\, \frac{1}{2}-x,\, \frac{3}{4}\right]$	[26,29]
15	$\left[-x,\, x+\frac{1}{2},\, \frac{3}{4}\right]$	[27,28]
16	$\left[x+\frac{1}{2},\, x,\, \frac{3}{4}\right]$	[30,32]

Table 12: Wyckoff site: 16l, site symmetry: ..2'

No.	position	mapping
1	$\left[x,\, x,\, \frac{1}{4}\right]$	[1,23]
2	$\left[-x,\, x,\, \frac{1}{4}\right]$	[2,21]
3	$\left[x,\, -x,\, \frac{1}{4}\right]$	[3,20]
4	$\left[x+\frac{1}{2},\, \frac{1}{2}-x,\, \frac{1}{4}\right]$	[4,19]
5	$\left[\frac{1}{2}-x,\, x+\frac{1}{2},\, \frac{1}{4}\right]$	[5,18]
6	$\left[-x,\, -x,\, \frac{1}{4}\right]$	[6,24]
7	$\left[x+\frac{1}{2},\, x+\frac{1}{2},\, \frac{1}{4}\right]$	[7,17]
8	$\left[\frac{1}{2}-x,\, \frac{1}{2}-x,\, \frac{1}{4}\right]$	[8,22]
9	$\left[-x,\, -x,\, \frac{3}{4}\right]$	[9,31]
10	$\left[x,\, -x,\, \frac{3}{4}\right]$	[10,29]
11	$\left[-x,\, x,\, \frac{3}{4}\right]$	[11,28]
12	$\left[\frac{1}{2}-x,\, x+\frac{1}{2},\, \frac{3}{4}\right]$	[12,27]
13	$\left[x+\frac{1}{2},\, \frac{1}{2}-x,\, \frac{3}{4}\right]$	[13,26]
14	$\left[x,\, x,\, \frac{3}{4}\right]$	[14,32]
15	$\left[\frac{1}{2}-x,\, \frac{1}{2}-x,\, \frac{3}{4}\right]$	[15,25]
16	$\left[x+\frac{1}{2},\, x+\frac{1}{2},\, \frac{3}{4}\right]$	[16,30]

Table 13: Wyckoff site: 16m, site symmetry: m..

No.	position	mapping
1	$[x,\, y,\, 0]$	[1,14]
2	$[-y,\, x,\, 0]$	[2,11]
3	$[y,\, -x,\, 0]$	[3,10]
4	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, \frac{1}{2}\right]$	[4,13]
5	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, \frac{1}{2}\right]$	[5,12]
6	$[-x,\, -y,\, 0]$	[6,9]
7	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, \frac{1}{2}\right]$	[7,16]
8	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, \frac{1}{2}\right]$	[8,15]
9	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, 0\right]$	[17,30]
10	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, 0\right]$	[18,27]
11	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, 0\right]$	[19,26]

continued ...

Table 13

No.	position	mapping
12	$\left[x,\, -y,\, \frac{1}{2}\right]$	[20,29]
13	$\left[-x,\, y,\, \frac{1}{2}\right]$	[21,28]
14	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, 0\right]$	[22,25]
15	$\left[y,\, x,\, \frac{1}{2}\right]$	[23,32]
16	$\left[-y,\, -x,\, \frac{1}{2}\right]$	[24,31]

Table 14: Wyckoff site: 32n, site symmetry: 1

No.	position	mapping
1	$[x,\, y,\, z]$	[1]
2	$[-y,\, x,\, z]$	[2]
3	$[y,\, -x,\, z]$	[3]
4	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$	[4]
5	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	[5]
6	$[-x,\, -y,\, z]$	[6]
7	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, \frac{1}{2}-z\right]$	[7]
8	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, \frac{1}{2}-z\right]$	[8]
9	$[-x,\, -y,\, -z]$	[9]
10	$[y,\, -x,\, -z]$	[10]
11	$[-y,\, x,\, -z]$	[11]
12	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[12]
13	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[13]
14	$[x,\, y,\, -z]$	[14]
15	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$	[15]
16	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$	[16]
17	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z\right]$	[17]
18	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, z\right]$	[18]
19	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, z\right]$	[19]
20	$\left[x,\, -y,\, \frac{1}{2}-z\right]$	[20]
21	$\left[-x,\, y,\, \frac{1}{2}-z\right]$	[21]
22	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z\right]$	[22]
23	$\left[y,\, x,\, \frac{1}{2}-z\right]$	[23]
24	$\left[-y,\, -x,\, \frac{1}{2}-z\right]$	[24]
25	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, -z\right]$	[25]
26	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, -z\right]$	[26]
27	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, -z\right]$	[27]
28	$\left[-x,\, y,\, z+\frac{1}{2}\right]$	[28]
29	$\left[x,\, -y,\, z+\frac{1}{2}\right]$	[29]
30	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, -z\right]$	[30]
31	$\left[-y,\, -x,\, z+\frac{1}{2}\right]$	[31]
32	$\left[y,\, x,\, z+\frac{1}{2}\right]$	[32]